

Presentation 2: Model Revisions

STAT 390 | Group 2 | Winter 2026

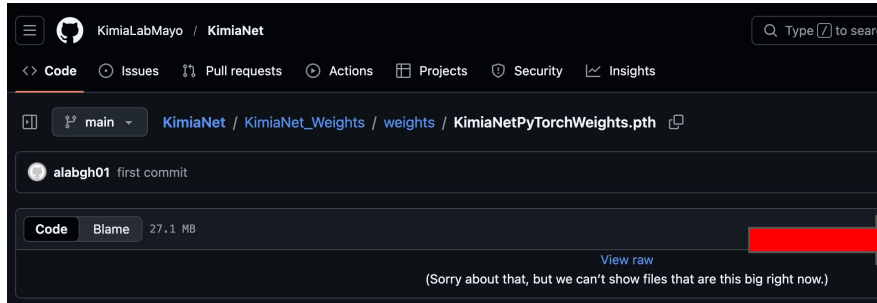
Adithi Adusumilli, Jason Chen, Roma Hashemi, Charlene Lu & Paige Usry

Part 1: Programmers

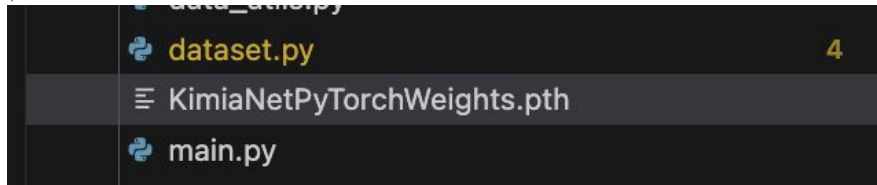
Mandatory Changes

KimiaNet

- Navigate to models.py
- Use KimiaNet instead of DenseNet as the pre-trained network
 - Reasoning: KimiaNet trained on histopathology images



Added file to our repository



Remove RandomResizedCrop()

- Navigate to dataset.py
- Use RandomResize() instead of RandomResizedCrop() as a data pre-processing step
 - To preserve aspect ratio and not distort cell / nucleus shape

```
if is_training:  
    # Training transforms with augmentation  
    transform = transforms.Compose([  
         transforms2.RandomResizedCrop(  
            min_size=int(0.9 * IMAGE_CONFIG['image_size'][0]),  
            max_size=int(1.1 * IMAGE_CONFIG['image_size'][0]),  
            transforms.Resize(IMAGE_CONFIG['image_size']),  
            transforms.RandomHorizontalFlip(),  
        ),  
    ])
```

- Purpose: Increase model accuracy

Remove Saturation Jitter

- Navigate to dataset.py
- When creating image transforms for training, remove the saturation=0.2 input from transforms.ColorJitter()

```
154 def create_transforms(is_training: bool = True) -> transforms.Compose:
155     """
156     Create image transforms for training or validation
157     """
158     if is_training:
159         # Training transforms with augmentation
160         transform = transforms.Compose([
161             # transforms.RandomResizedCrop(IMAGE_CONFIG['image_size'][0]),
162             transforms.RandomHorizontalFlip(),
163             transforms.RandomVerticalFlip(),
164             transforms.RandomRotation(15),
165             transforms.ColorJitter(brightness=0.2, contrast=0.2), # removed , saturation=0.2
166             transforms.ToTensor(),
```

- Reasoning: increases model accuracy

Use AdamW

- Navigate to dataset.py
- Replace `optim.Adam(...)` with `optim.AdamW(...)` when initializing optimizer



```
27 # Initialize optimizer and criterion
28 self.optimizer = optim.AdamW(
29     self.model.parameters(),
30     lr=TRAINING_CONFIG['learning_rate'],
31     weight_decay=TRAINING_CONFIG['weight_decay']
32 )
```

- Reasoning: AdamW can achieve better convergence of the loss function and is more likely to find global minimum instead of local minimum

Use persistent_workers=True

- Navigate to main.py
- When creating data loaders for train/val/test splits, include **persistant_workers=True** as an input to `DataLoader()`
- Reasoning: Reduces CPU "dead time" between epochs

```
197     # Create data loaders
198     train_loader = DataLoader(
199         train_ds, batch_size=args.batch_size, shuffle=True,
200         num_workers=args.num_workers, pin_memory=True, collate_fn=case_collate_fn,
201         persistent_workers=True
202     )
203
204     val_loader = DataLoader(
205         val_ds, batch_size=args.batch_size, shuffle=False,
206         num_workers=args.num_workers, pin_memory=True, collate_fn=case_collate_fn,
207         persistent_workers=True
208     )
209
210     test_loader = DataLoader(
211         test_ds, batch_size=args.batch_size, shuffle=False,
212         num_workers=args.num_workers, pin_memory=True, collate_fn=case_collate_fn,
213         persistent_workers=True
214     )
215
```



Optuna Hyperparameter Training

Difficult Task

Why Optuna?

- Grid search or random search would be impractical given the size of the search space
- Optuna (Bayesian Optimization) provides an efficient trade-off between optimal parameter search and training speed
 - Enables identification of ideal hyperparameter values without excess runtime
- Finding efficient hyperparameter values can **increase the model accuracy**

Code additions and changes mostly made to a copy of main.py, which we will describe. Small edits made to copies of trainer.py and models.py to support the hyperparameter tuning with Optuna.

Hyperparameters Tuned

- Architecture
 - embed_dim: [128, 256, 512]
 - attention_hidden_dim: [64, 128, 256]
 - dropout: (0.1, 0.5)
- Training
 - learning_rate: (1e-5, 1e-2) – log
 - weight_decay: (1e-6, 1e-2) – log
 - epochs: (5, 20)
 - batch_size: [1, 2, 4]
- Class Weights
 - Tuned vs. fixed

Value ranges were assigned near and based on the model's previous fixed values, referenced Team 6 Presentation 1 Mathematical Operations section

ModelHyperparams

- Create a class definition for the model's hyperparameters
 - Makes the pipeline cleaner and easier to scale

```
class ModelHyperparameters:
    def __init__(self, embed_dim, attention_hidden_dim, epochs, lr, batch_size,
                 dropout=0.2, weight_decay=0.0, class_weights=[1,1,1]):
        # Architecture
        self.embed_dim = embed_dim
        self.attention_hidden_dim = attention_hidden_dim
        self.per_slice_cap = 100
        self.max_slices_per_stain = 10
        self.dropout = dropout
        # Training
        self.learning_rate = lr
        self.weight_decay = weight_decay
        self.epochs = epochs
        self.batch_size = batch_size
        self.class_weights = class_weights
```


choose_parameters()

- Defines the search space and generates hyperparameter configurations

```
def choose_parameters(trial):  
    # Architecture  
    embed_dim = trial.suggest_categorical('embed_dim', [128, 256, 512])  
    attention_hidden_dim = trial.suggest_categorical('attention_hidden_dim', [64, 128, 256])  
    dropout = trial.suggest_float('dropout', 0.1, 0.5)  
    per_slice_cap = 100  
    max_slices_per_stain = 10  
    # Training  
    learning_rate = trial.suggest_float('learning_rate', 1e-5, 1e-2, log=True)  
    weight_decay = trial.suggest_float('weight_decay', 1e-6, 1e-2, log=True)  
    epochs = trial.suggest_int('epochs', 5, 20)  
    batch_size = trial.suggest_categorical('batch_size', [1, 2, 4])  
    # Class weights  
    class_weights = [  
        trial.suggest_float('w1', 0.5, 2.0),  
        trial.suggest_float('w2', 0.5, 2.0),  
        trial.suggest_float('w3', 0.5, 2.0)  
    ]  
  
    return ModelHyperparameters(  
        embed_dim=embed_dim,  
        attention_hidden_dim=attention_hidden_dim,  
        epochs=epochs,  
        lr=learning_rate,  
        batch_size=batch_size,  
        dropout=dropout,  
        weight_decay=weight_decay,  
        class_weights=class_weights  
    )
```

objective()

- Defines what is being optimized- the validation loss for one trial (Binary Cross Entropy loss pulled from the trainer)
- Samples a set of hyperparameters (learning rate, dropout, embedding size, attention size)
- Builds and trains MIL model for a few epochs (3 here) using those parameters
- Evaluates validation loss after each epoch
- Uses pruning to stop bad trials early for efficiency

```
def objective(trial, train_loader, val_loader, device):  
    params = choose_parameters(trial)  
    model = create_model(  
        num_classes=MODEL_CONFIG['num_classes'],  
        embed_dim=params.embed_dim,  
        attention_hidden_dim=params.attention_hidden_dim,  
        dropout=params.dropout  
    )  
    trainer = MILTrainer2(  
        model=model,  
        device=device,  
        learning_rate=params.learning_rate,  
        weight_decay=params.weight_decay,  
        checkpoint_dir=None  
    )  
    epochs = 3 #small for first run  
    best_val_loss = float("inf")
```

```
    for epoch in range(epochs):  
        trainer.train_epoch(train_loader)  
        val_loss, _ = trainer.validate(val_loader)  
  
        trial.report(val_loss, epoch)  
  
        if trial.should_prune():  
            raise optuna.TrialPruned()  
  
        if val_loss < best_val_loss:  
            best_val_loss = val_loss  
  
    return best_val_loss
```

main()

- Previously ran full model training but restructured to run hyperparam optimization now
- Creates the optuna study with validation loss being minimized and pruning enabled
- Executes multiple trials (7), each training for up to 3 epochs
- Reports the final best validation loss and corresponding hyperparameters

```
def main():
    args = parse_args()

    set_seed(args.seed)
    device = get_device()

    train_data, val_data, test_data = prepare_data(args)

    train_loader, val_loader, _ = create_data_loaders(
        train_data, val_data, test_data, args
    )
    study = optuna.create_study(direction="minimize", pruner=optuna.pruners.MedianPruner(n_startup_trials=2))

    study.optimize(
        lambda trial: objective(trial, train_loader, val_loader, device),
        n_trials=7
    )
```

```
print("\n" + "=" * 60)
print("OPTUNA FINISHED")
print("=" * 60)

print(f"Best Validation Loss: {study.best_value}")
print("Best Hyperparameters:")
for key, value in study.best_params.items():
    print(f"  {key}: {value}")

if __name__ == "__main__":
    main()
```

Tuning Results

```
=====
OPTUNA FINISHED
=====
```

```
Best Validation Loss: 0.48865219753454714
```

```
Best Hyperparameters:
```

```
  embed_dim: 512
```

```
  attention_hidden_dim: 128
```

```
  dropout: 0.11750676059787302
```

```
  learning_rate: 0.00023730809517398077
```

```
  weight_decay: 0.0023655471180361997
```

```
  epochs: 10
```

```
  batch_size: 2
```

```
  w1: 1.013814165434012
```

```
  w2: 0.8248287308051502
```

```
  w3: 0.6899884408834002
```

Next Steps

- Expand the Optuna study (more trials and epochs) for more complete search
- Refine the parameter search space based on initial results
- Retrain the model using these best hyperparams to evaluate the effect on model performance
 - Analyze / experiment with specifically the optimized dropout values across the three usages (ex. 0.11 vs. previous 0.3) to assess impact

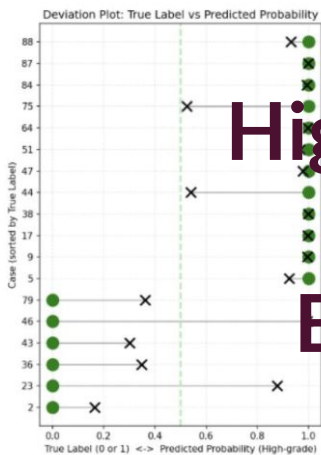
Part 2: Biological Analysis

Adithi & Jason

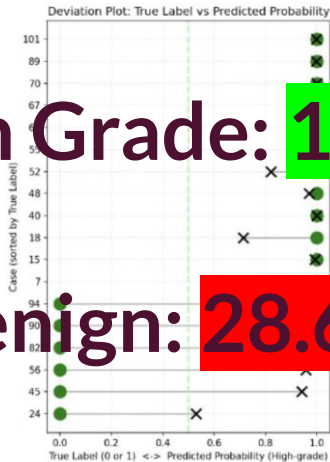
2.1 Fall 2025 Test Results

High Grade: 100% Correct Classification

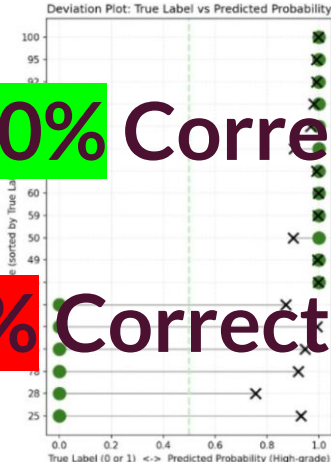
Benign: 28.6% Correct Classification



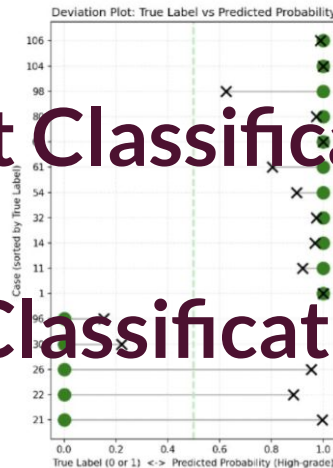
split_01



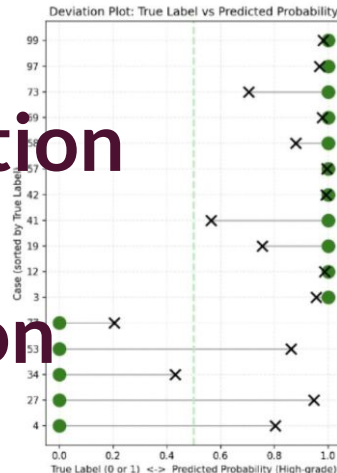
split_02



split_03



split_04



split_05

High: 12/12

Benign: 4/6

High: 12/12

Benign: 0/6

High: 12/12

Benign: 0/6

High: 11/11

Benign: 2/5

High: 11/11

Benign: 2/5

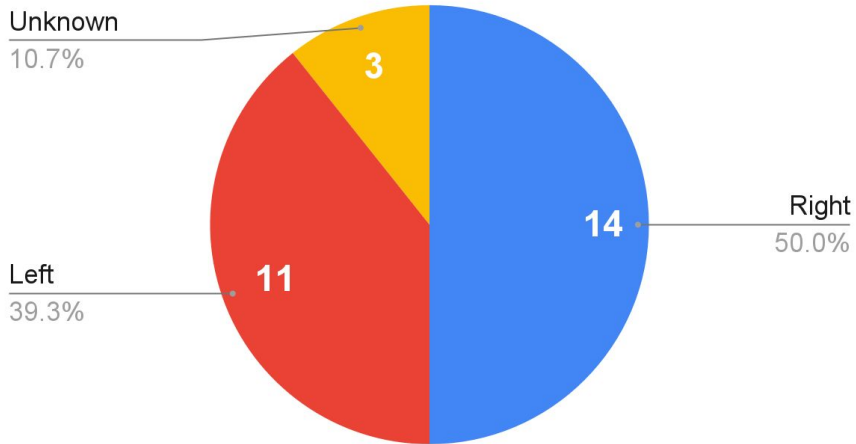
2.2 Demographic Breakdown of Results

Left: 18.2% Correct Classification
Right: 35.7% Correct Classification
Unknown: 33.3% Correct Classification

Laterality refers to which eye was impacted

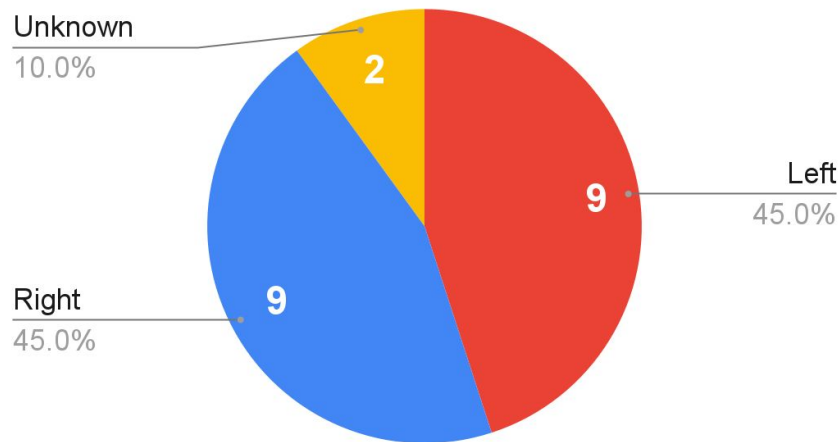
All Benign Cases:

Laterality



Misclassified Benign Cases:

Laterality

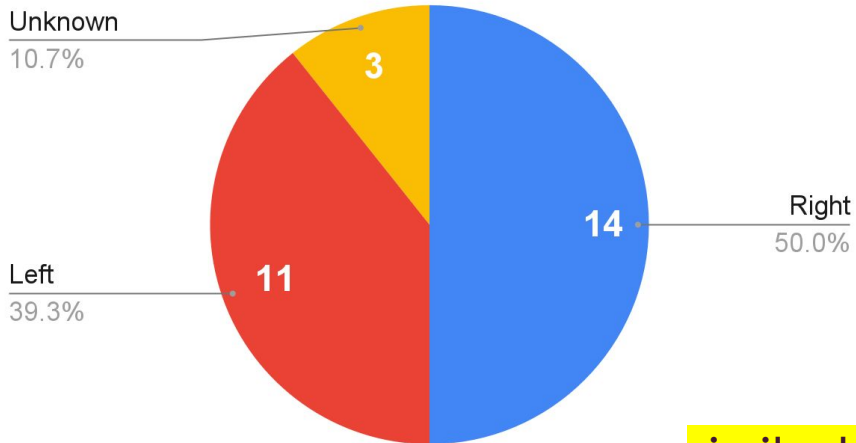


2.2 Demographic Breakdown of Results

Laterality - refers to which eye was impacted

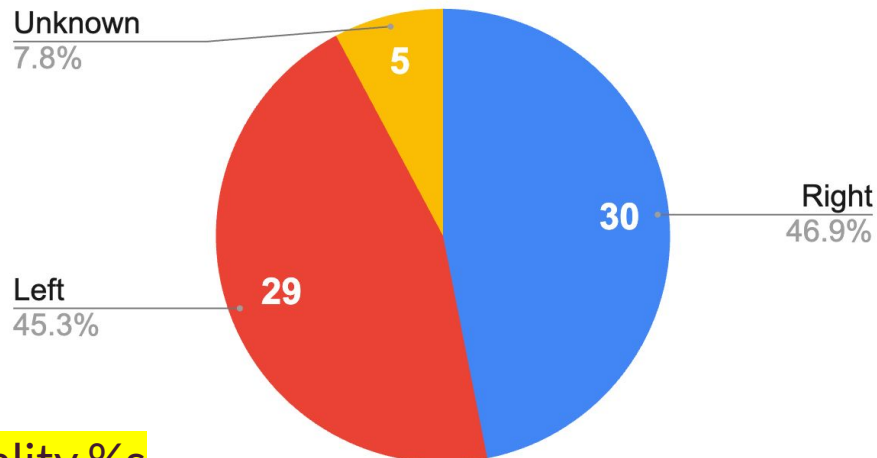
All Benign Cases:

Laterality



All High-Grade Cases:

Laterality



similar laterality %s

2.2 Demographic Breakdown of Results

20-35: 25% Correct Classification

35-50: 16.7% Correct Classification

50-65: 40% Correct Classification

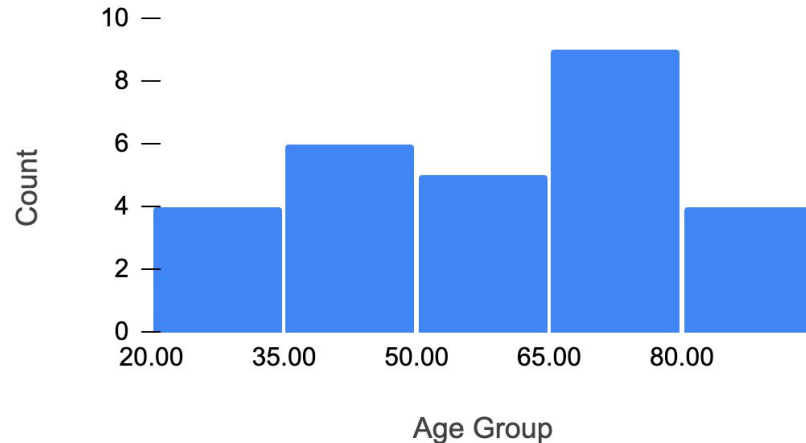
65-80: 33.3% Correct Classification

80+: 25% Correct Classification

Age

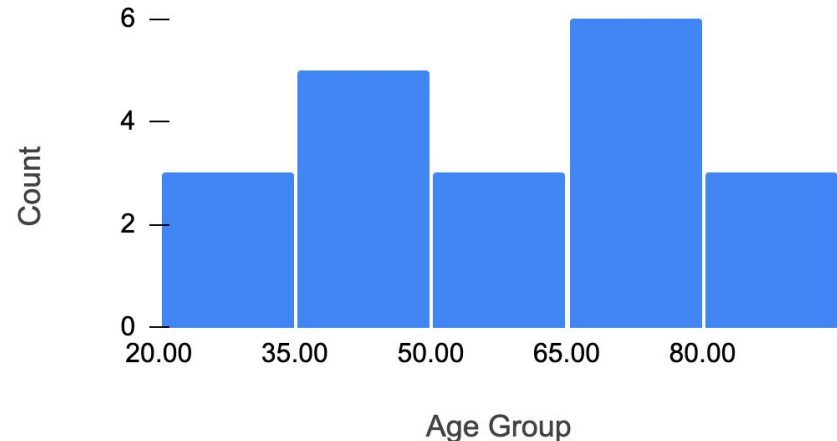
All Benign Cases:

Age Distribution



Misclassified Benign Cases:

Age Distribution

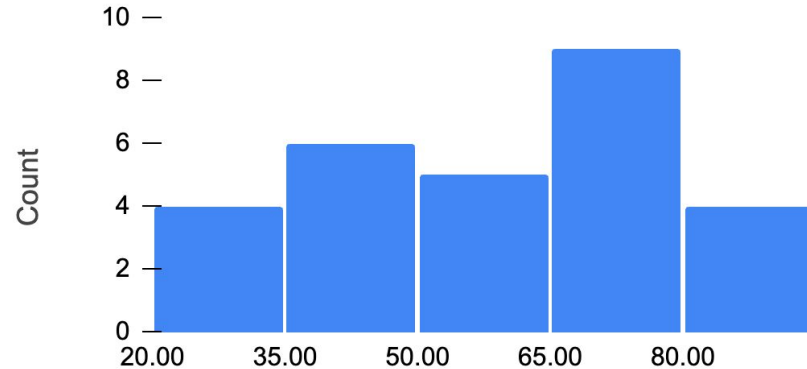


2.2 Demographic Breakdown of Results

Age

All Benign Cases:

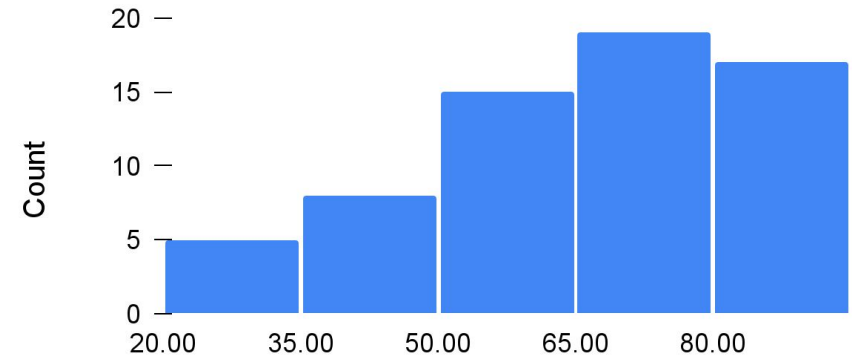
Age Distribution



no major skew

All High-Grade Cases:

Age Distribution



skewed left!

Male: 45.5% Correct Classification

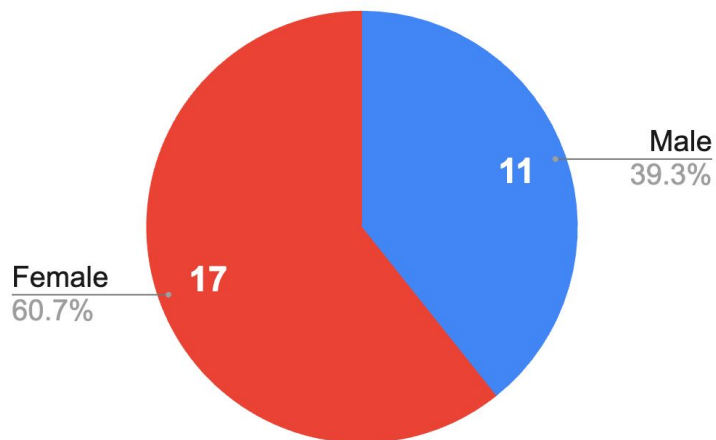
2.2 Demographic Breakdown of Results

Female: **17.6%** Correct Classification

Biological Sex

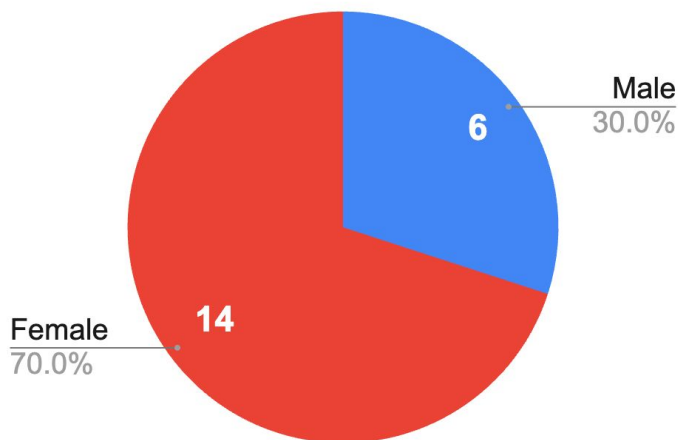
All Benign Cases:

Biological Sex



Misclassified Benign Cases:

Biological Sex

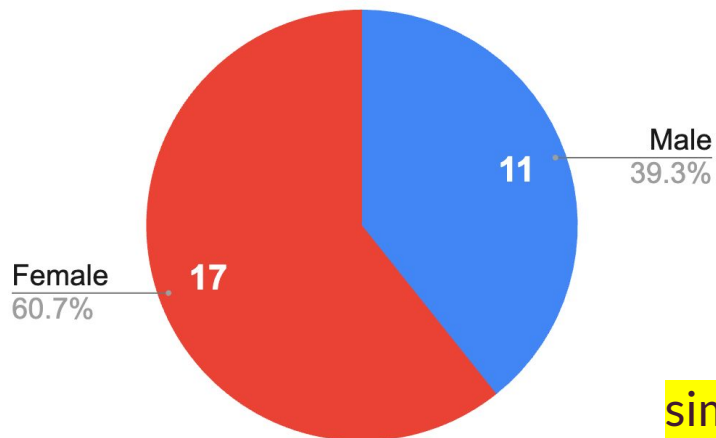


2.2 Demographic Breakdown of Results

Biological Sex

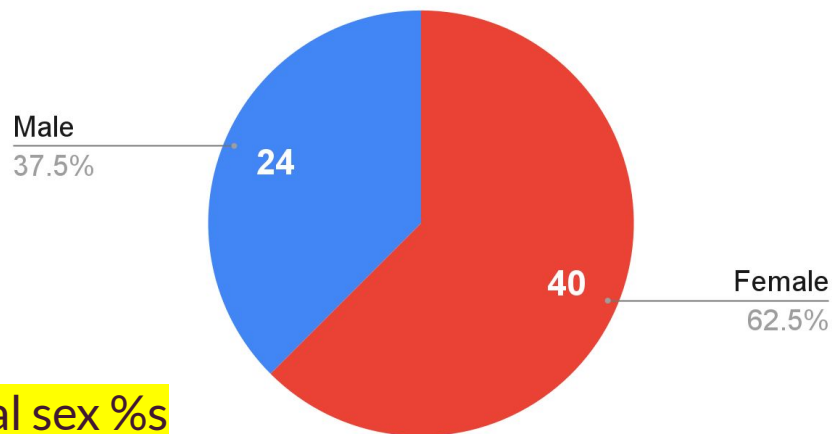
All Benign Cases:

Biological Sex



All High-Grade Cases:

Biological Sex



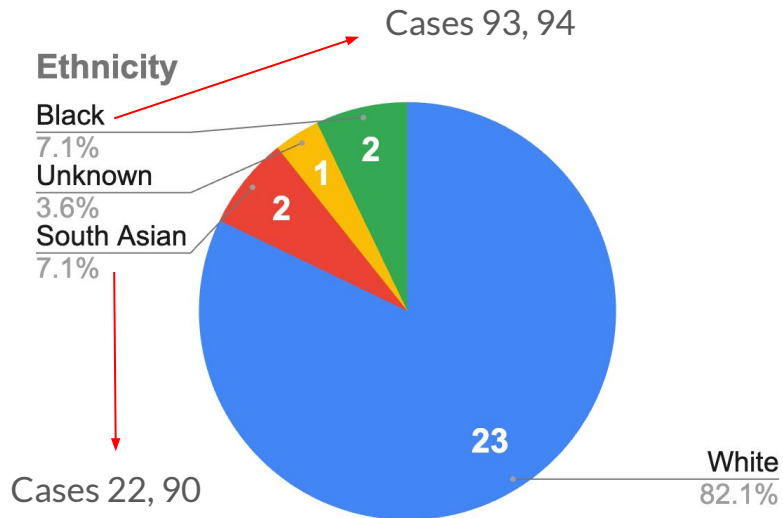
similar biological sex %s

2.2 Demographic Breakdown of Results

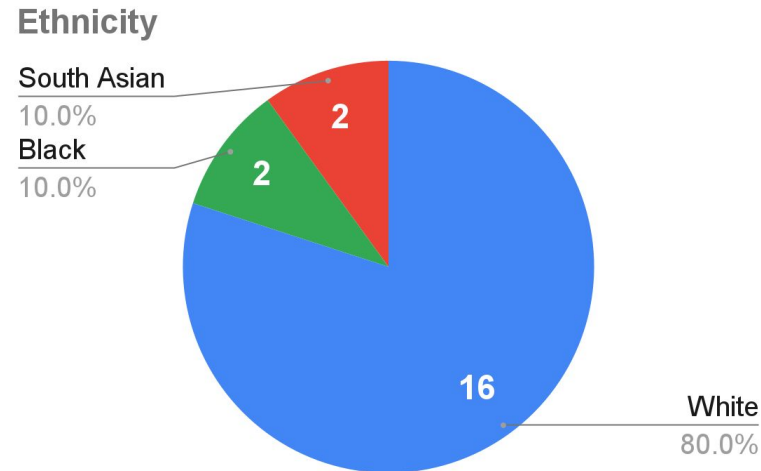
Black: 0% Correct Classification
South Asian: 0% Correct Classification
White: 30.4% Correct Classification

Ethnicity

All Benign Cases:



Misclassified Benign Cases:

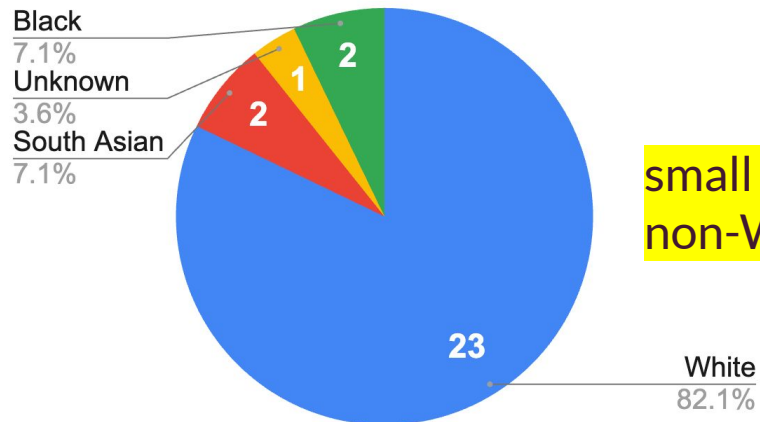


2.2 Demographic Breakdown of Results

Ethnicity

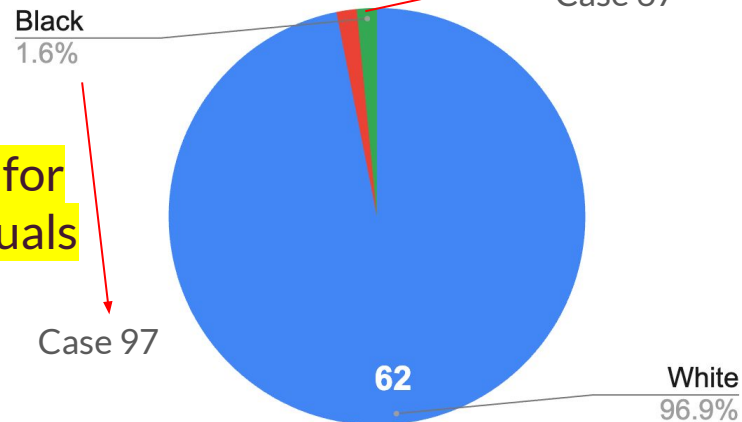
All Benign Cases:

Ethnicity



All High-Grade Cases:

Ethnicity



small sample sizes for
non-White individuals

2.2 Demographic Breakdown of Results

Caruncle: 0% Correct Classification

Hornix: 100% Correct Classification

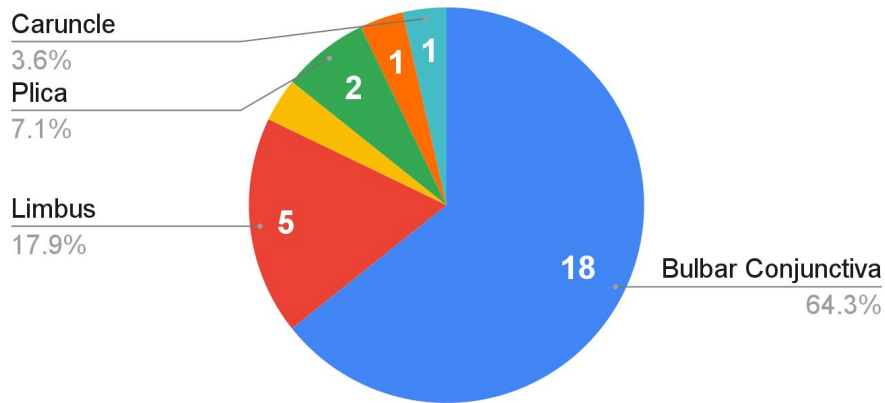
Plica: 0% Correct Classification

Limbus: 20% Correct Classification

Lesion Location Bulbar Conjunctiva: 33.3% Correct Classification

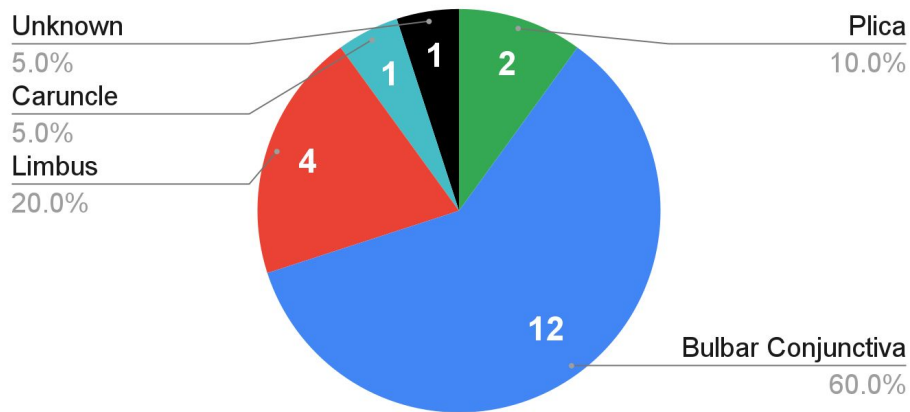
All Benign Cases:

Lesion Location



Misclassified Benign Cases:

Lesion Location



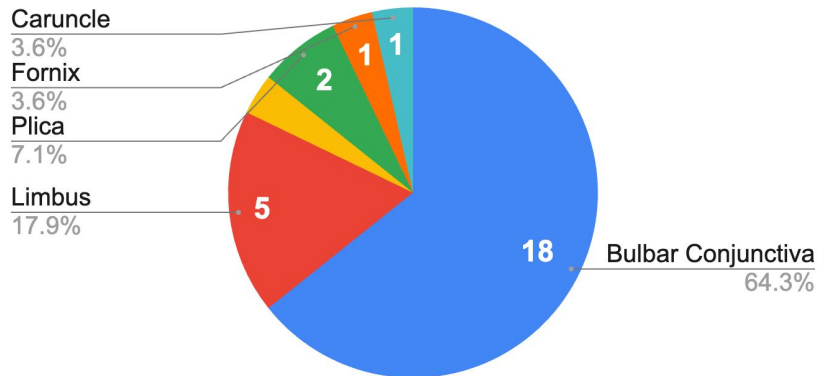
2.2 Demographic Results

Comparatively higher %
high-grade lesions in Limbus,
lower % in Bulbar
Conjunctiva

Lesion Location

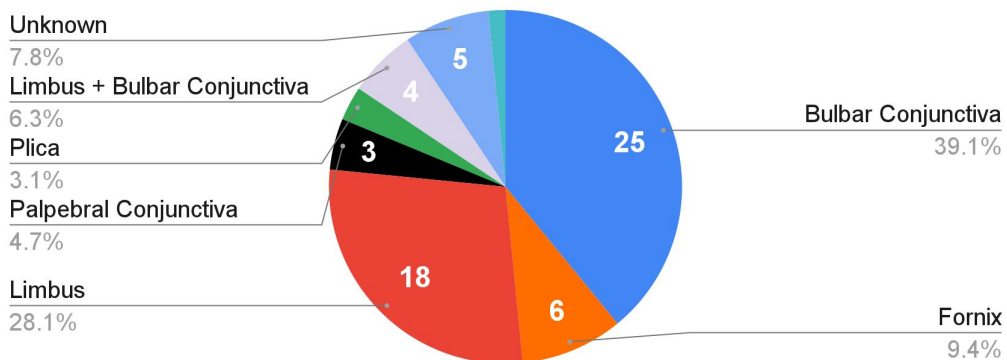
All Benign Cases:

Lesion Location

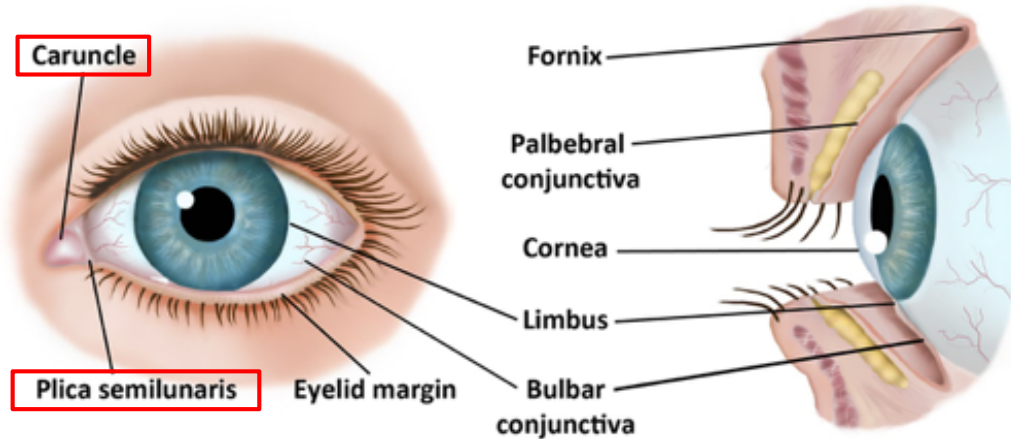


All High-Grade Cases:

Lesion Location

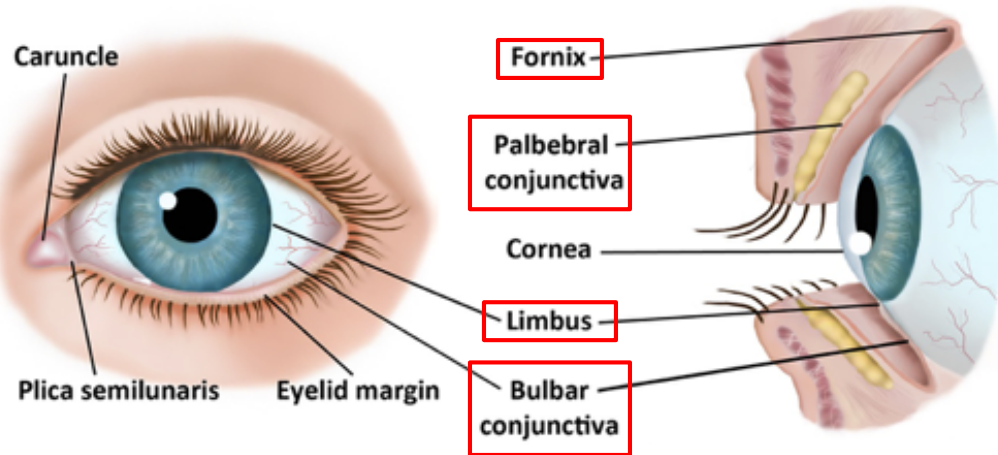


2.3 Anatomy of Conjunctiva



- **Plica** and **caruncle** are two regions of the conjunctival epithelium aiding in eye lubrication, tear drainage, and eye movement
- Literature shows that they have an abundance of *goblet cells* → cells vital for eye lubrication and tear production
 - Our histological slices show higher density of goblet cells than fornix, limbus, bulbar conjunctiva, or palpebral conjunctiva
- Can contribute to model misclassification in these 2 regions

2.3 Anatomy of Conjunctiva (cont.)



Bulbar Conjunctiva

- Protective covering for anterior sclera (white of eye)

Fornix

- Junction between palpebral and bulbar conjunctival regions
- Contributes to tear production

Palpebral Conjunctiva

- Lines inner eyelid surface for protection and facilitation of eyelid movement

Limbus

- Border region where conjunctiva meets cornea
- Barrier against conjunctival cell growth into cornea

2.4 Conditions of Cells in Conjunctiva

Goblet Cell Hyperplasia

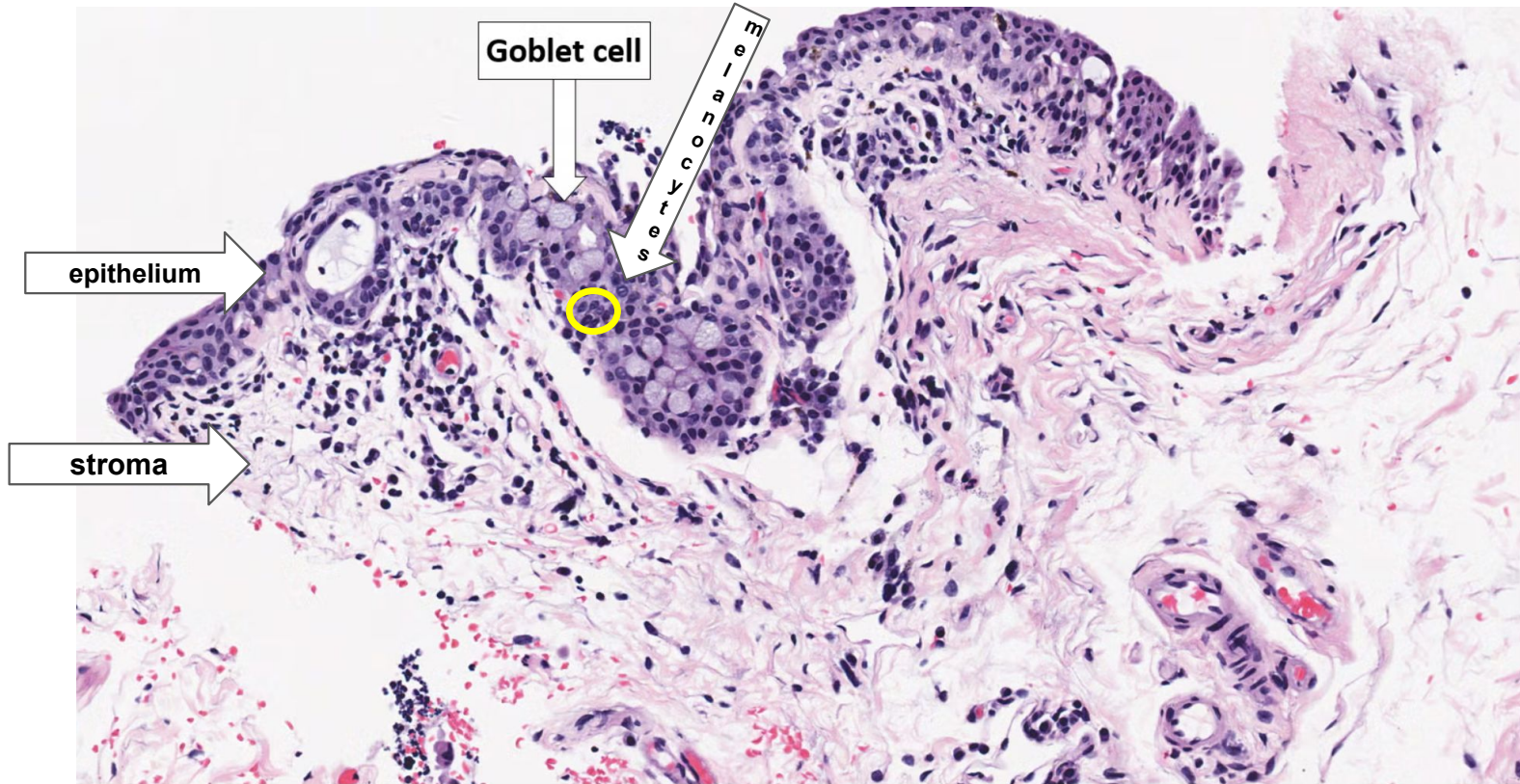
- Proliferation/Increase in number of goblet cells
- **Seen in all of the benign plica & caruncle slices**
- **Benign condition!**
 - As model is trained primarily on bulbar conjunctiva and limbus, this could be contributing to misclassification of goblet cell-rich regions

Melanosis

- Increase in number of melanocytes
- **Often malignant/type of C-MIL**

Hypothesis: model could be misconstruing goblet cell hyperplasia for melanosis, which may be leading to misclassification of benign slices

2.5 Conjunctival Intraepithelial Space Histology

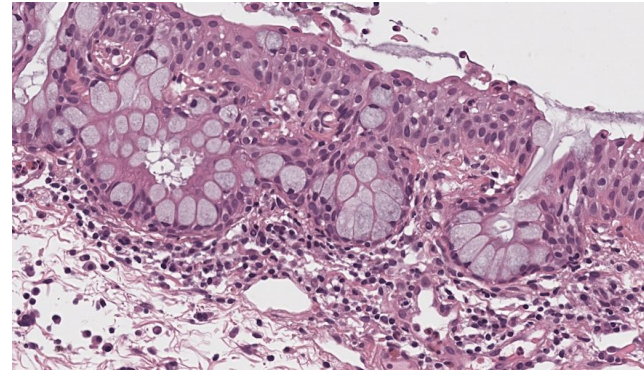
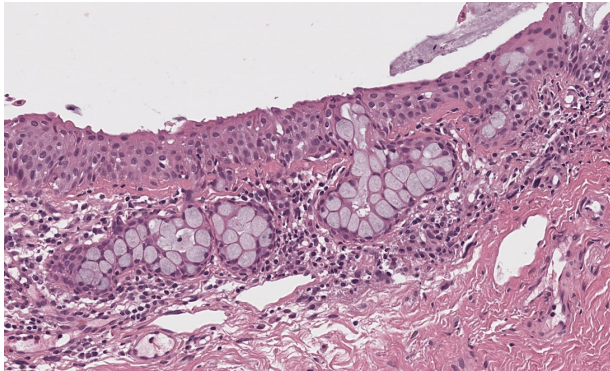
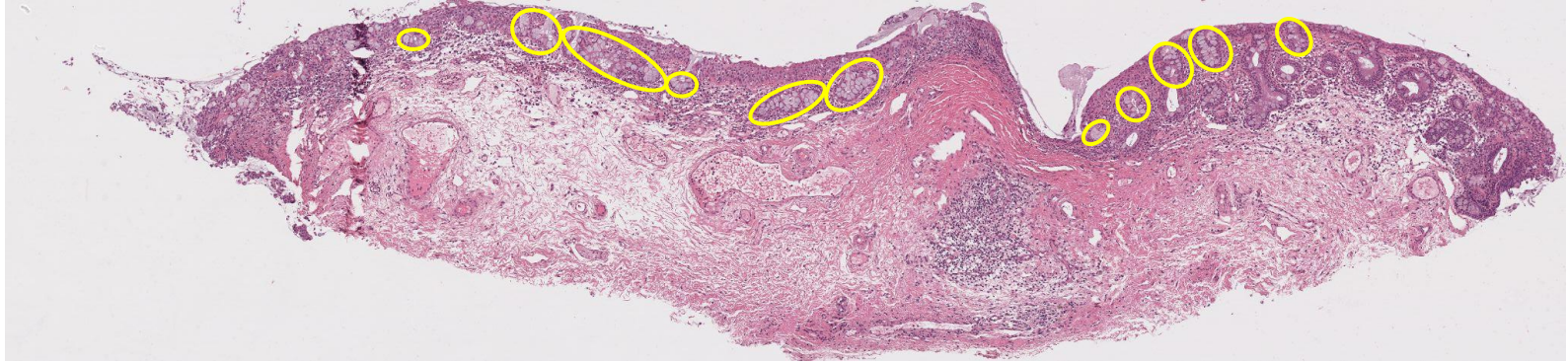


Location: Plica

Case 46 - benign, **misclassified**

39 year old White male

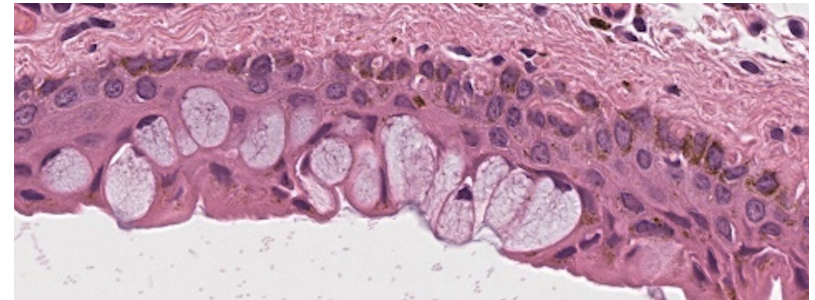
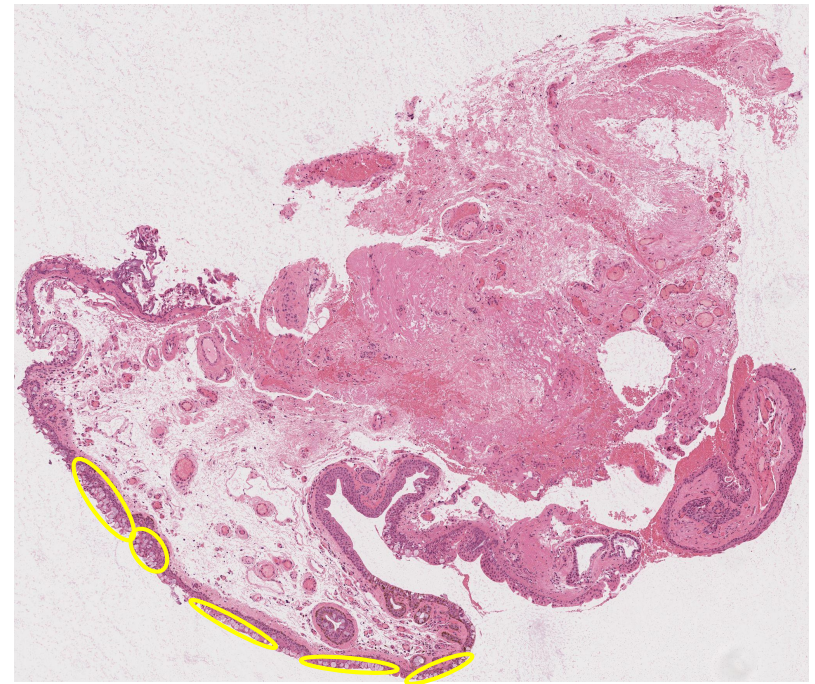
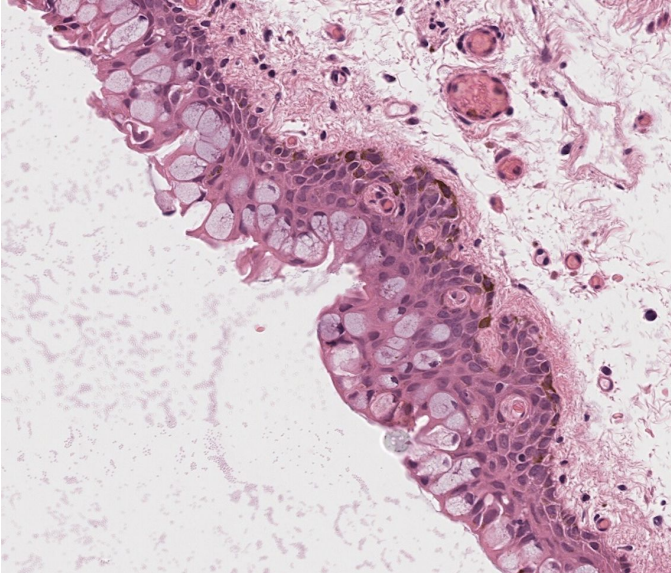
Finding: Many goblet cell bulks throughout
intraepithelial space



Location: Plica

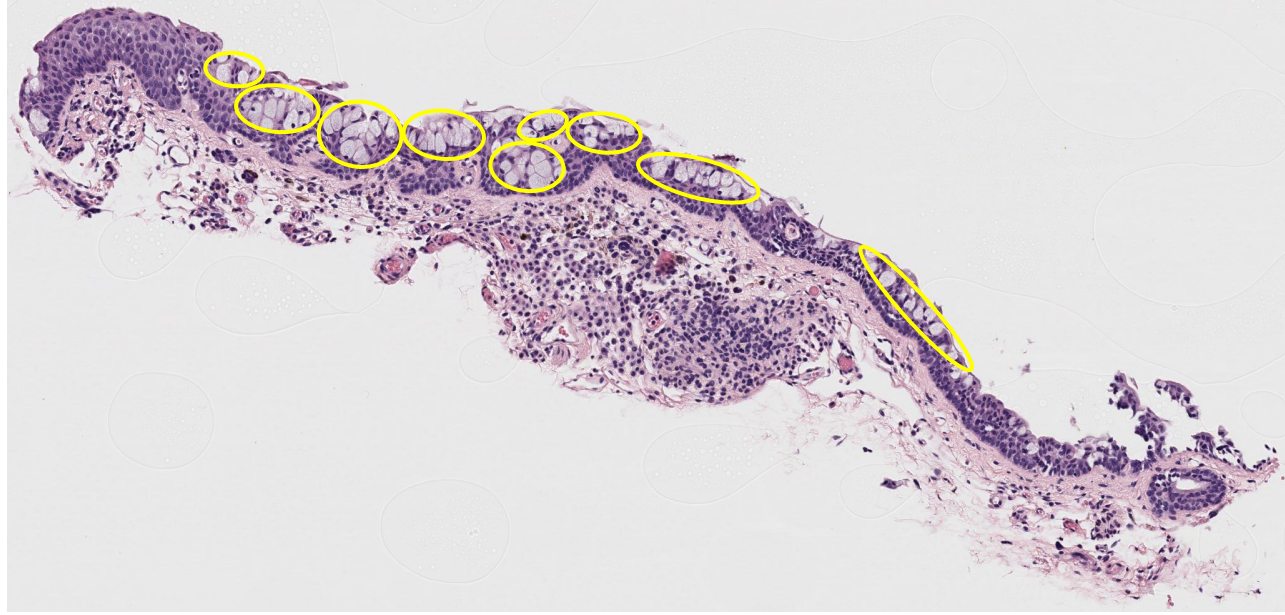
Case 86 - benign, **misclassified**
78 year old White female

Finding: Many goblet cell bulks
throughout intraepithelial region

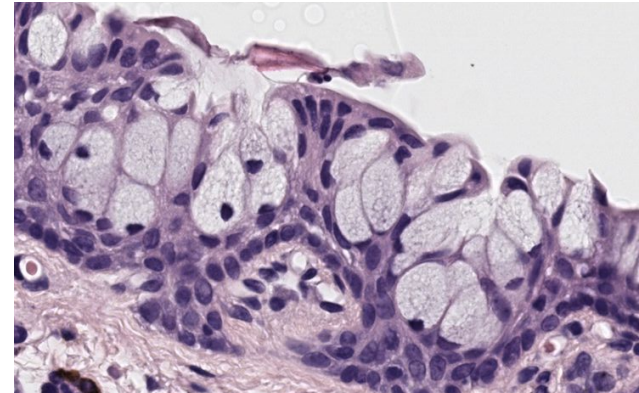
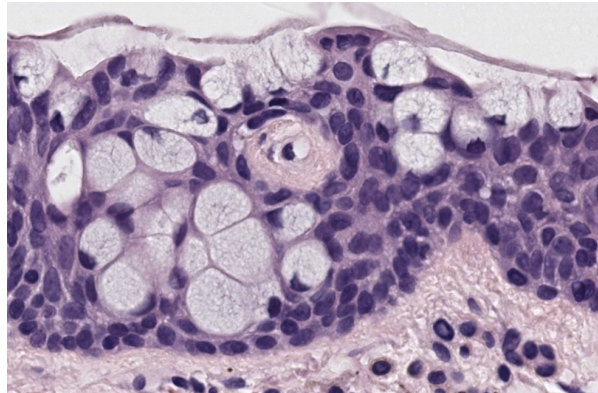


Location: Plica

Case 20 - high-grade (3)
86 year old White male



Finding: Very large goblet cell bulks throughout intraepithelial space



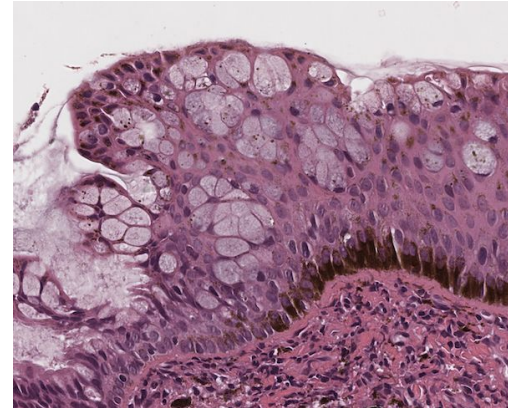
Location: Caruncle

Case 90 - benign,

misclassified

24 year old South Asian
female

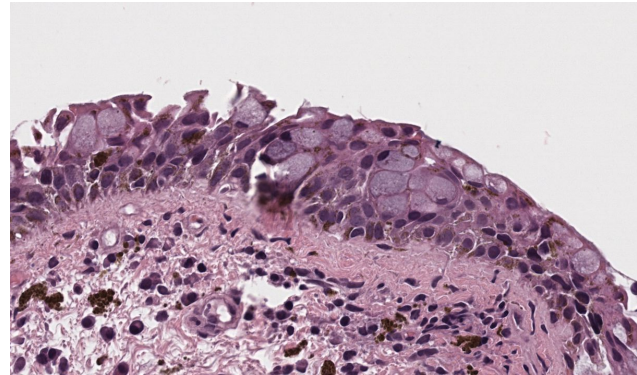
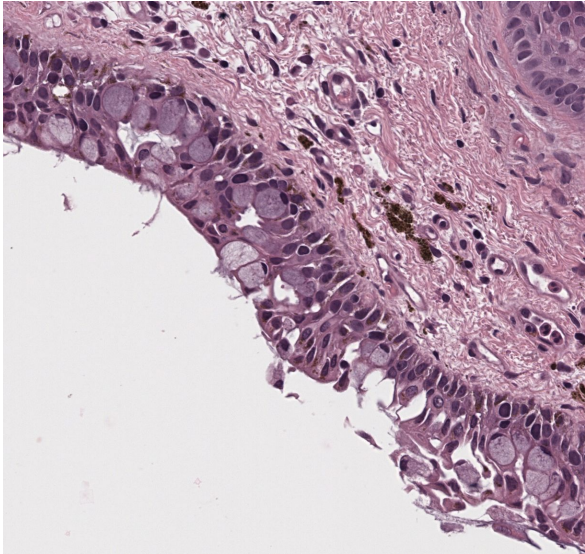
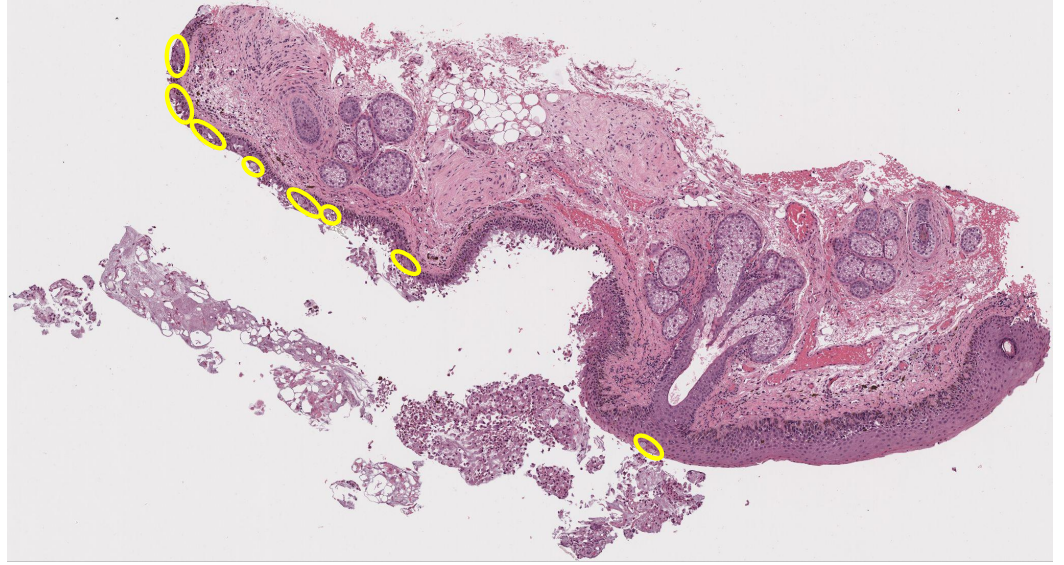
Finding: Many goblet
clumps throughout



Location: Caruncle

Case 102 - high-grade (3)
71 year old White female

Finding: Many goblet cell
bulks throughout
intraepithelial space

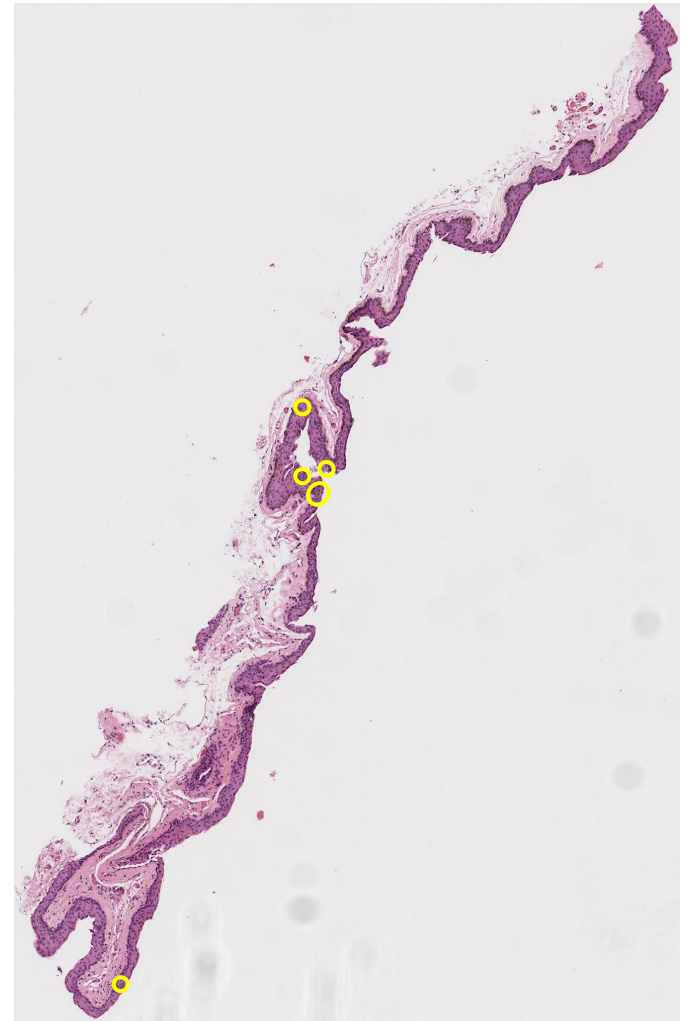
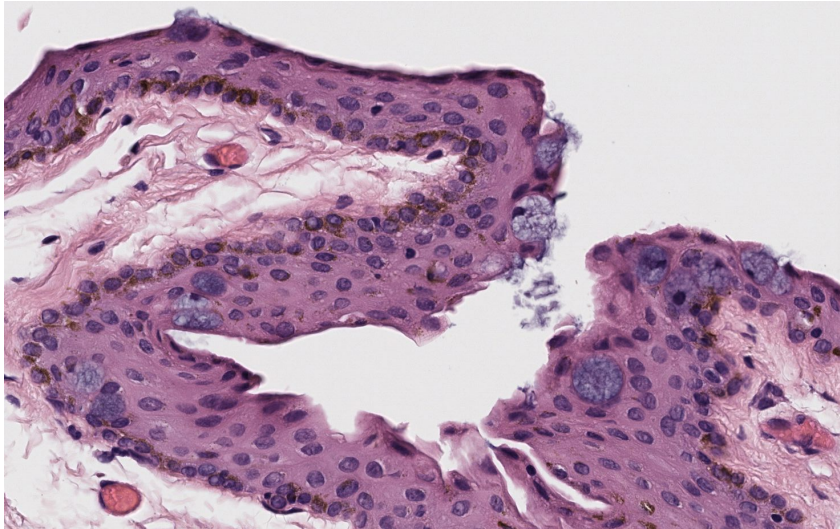


Location: Bulbar Conjunctiva

Case 30 - benign

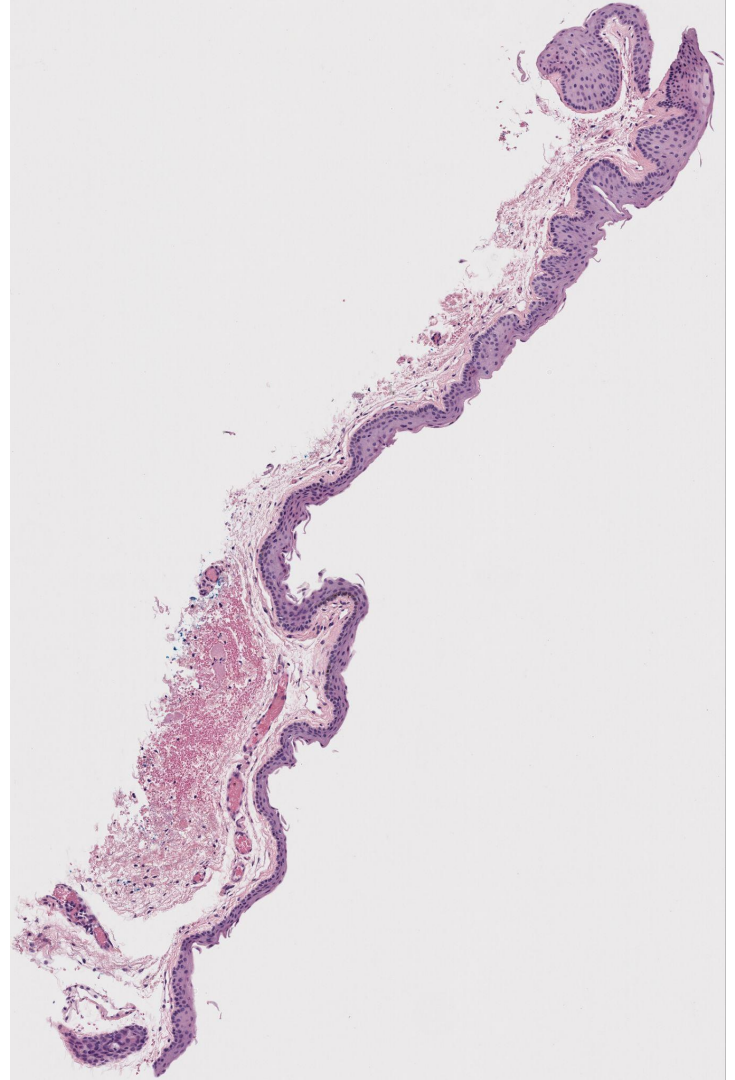
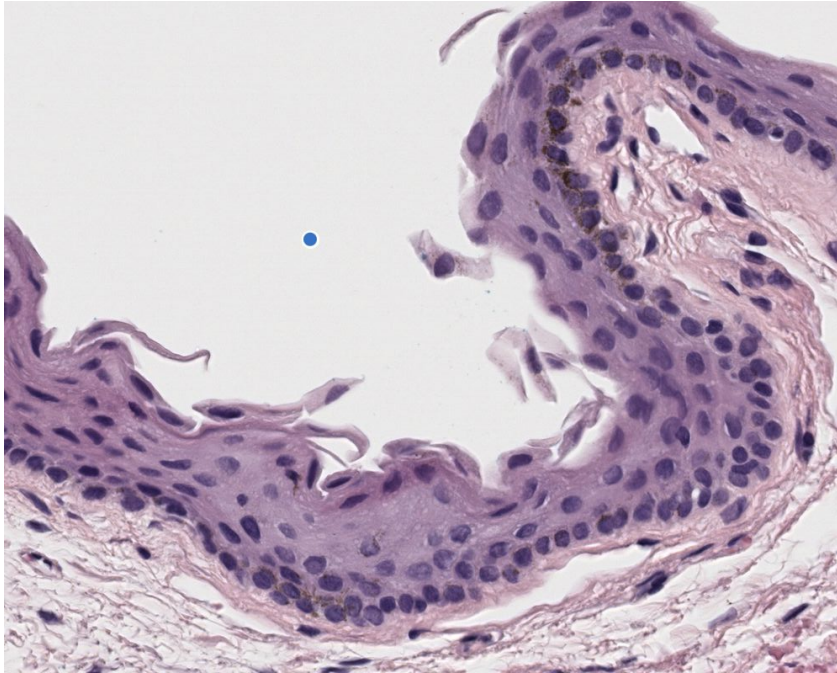
86 year old White female

Finding: Some smaller individual goblet cell
bunks throughout intraepithelial space (but
not as much as plica/caruncle region)



Location: Bulbar Conjunctiva
Case 26 - benign, **misclassified**
70 year old White male

Finding: No goblet cell bulks in intraepithelial space



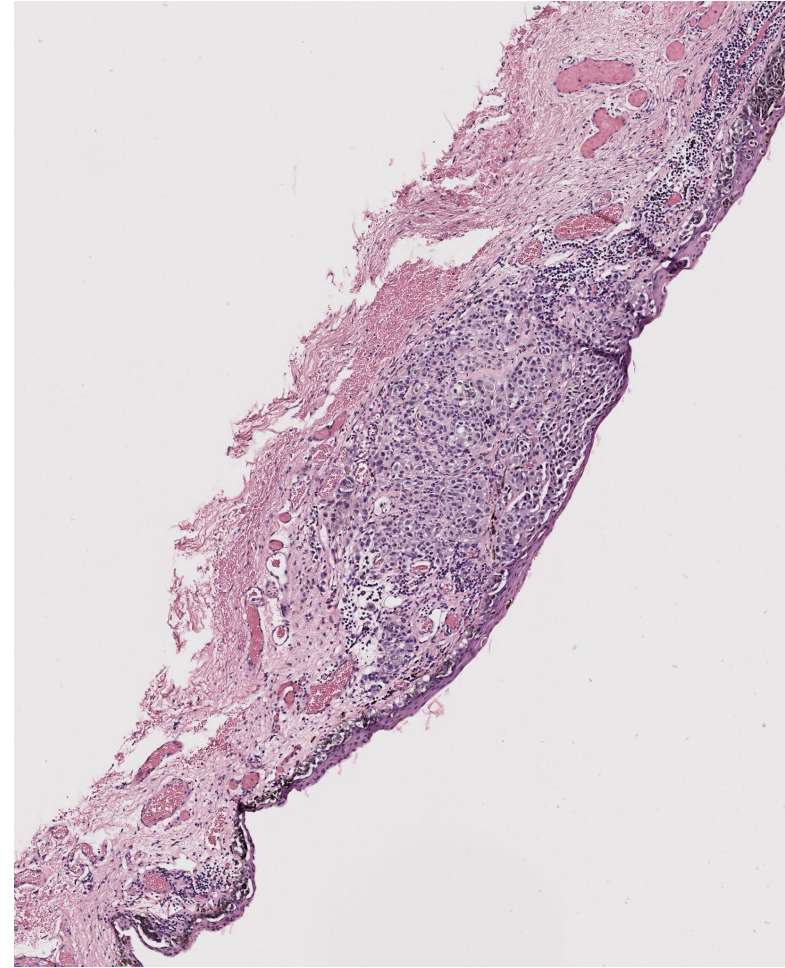
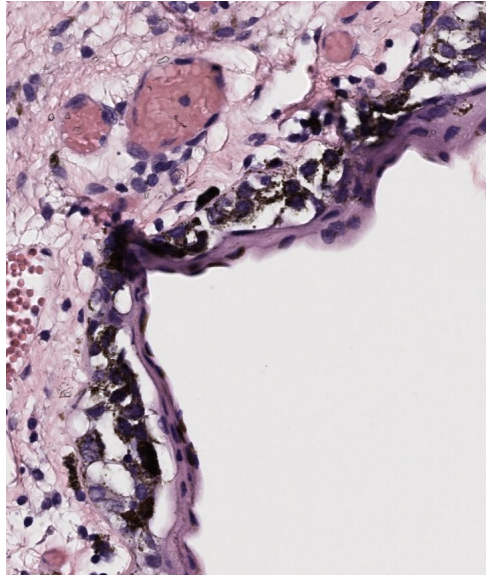
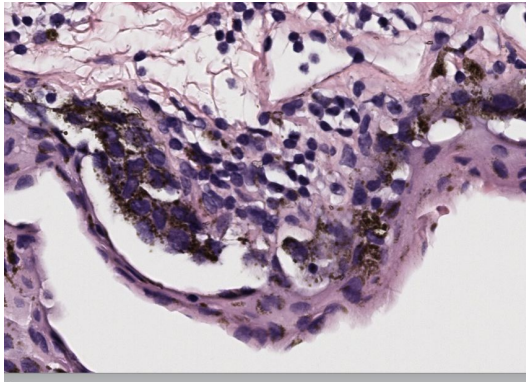
Location: Bulbar Conjunctiva

Case 14 - high-grade (4)

55 year old White male

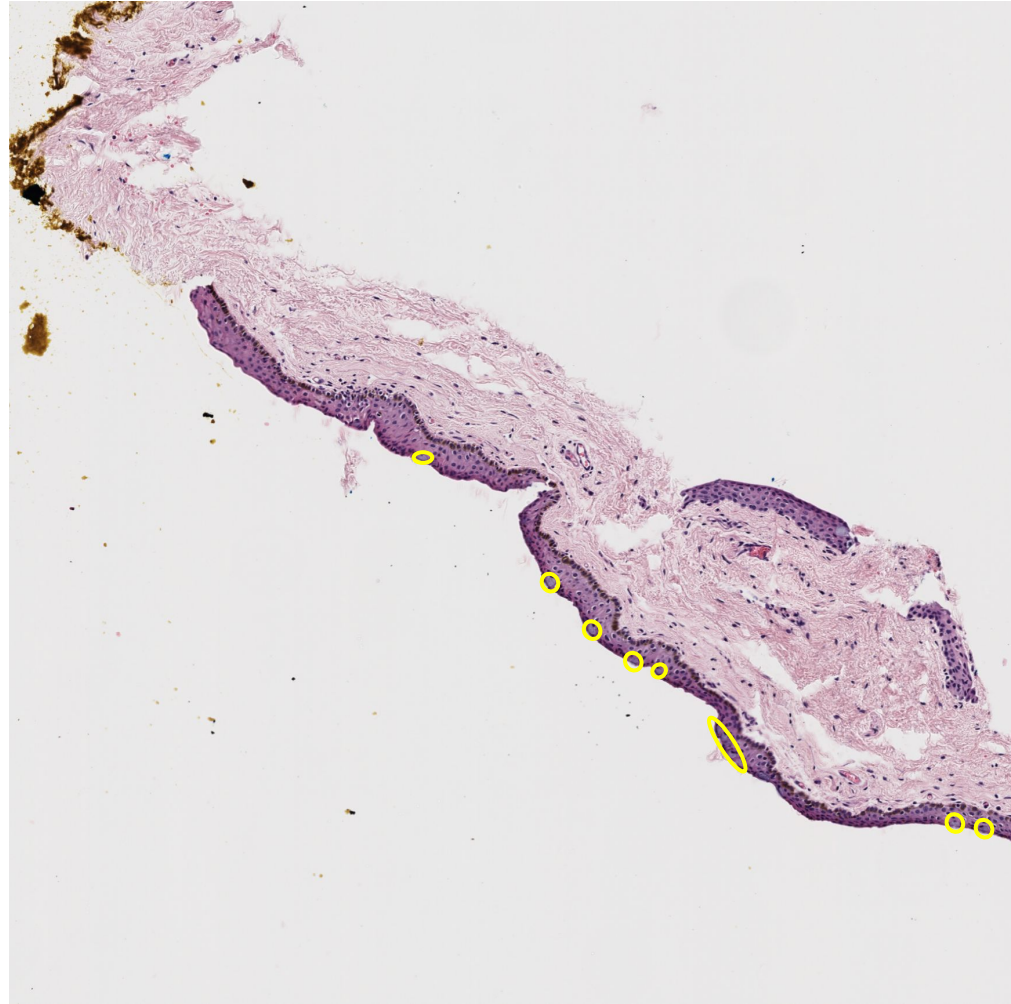
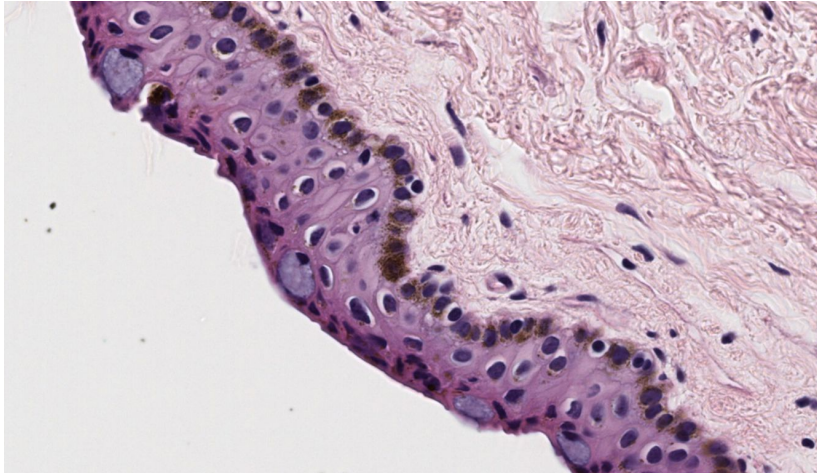
Finding: No goblet cell bulks in
intraepithelial space

- Clear melanoma



Location: Limbus
Case 36 - benign
40 year old White female

Finding: Some smaller individual goblet cell bulks throughout intraepithelial space (but not as much as plica/caruncle region)

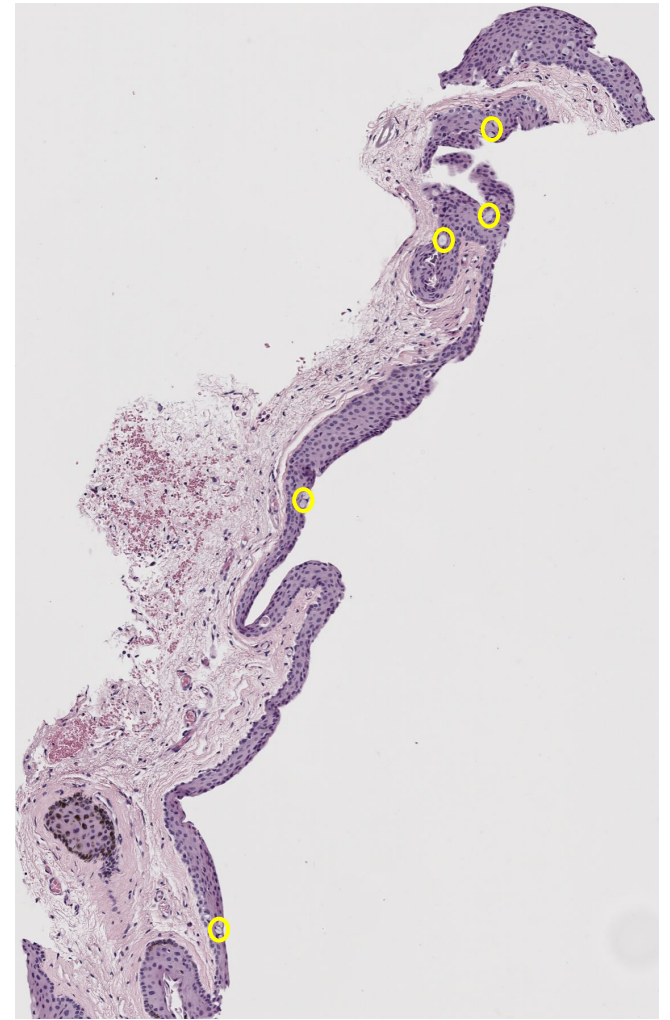
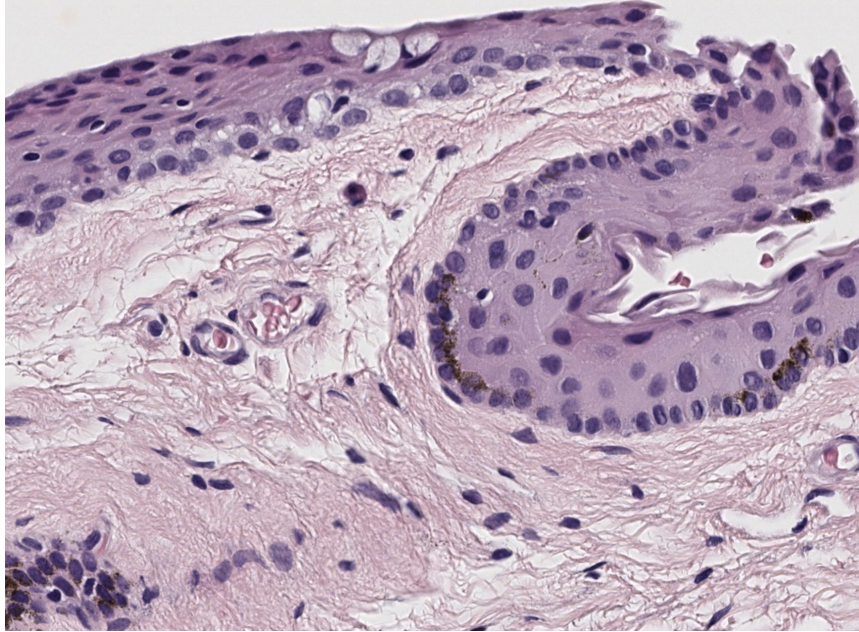


Location: Limbus

Case 24 - benign, **misclassified**

74 year old White female

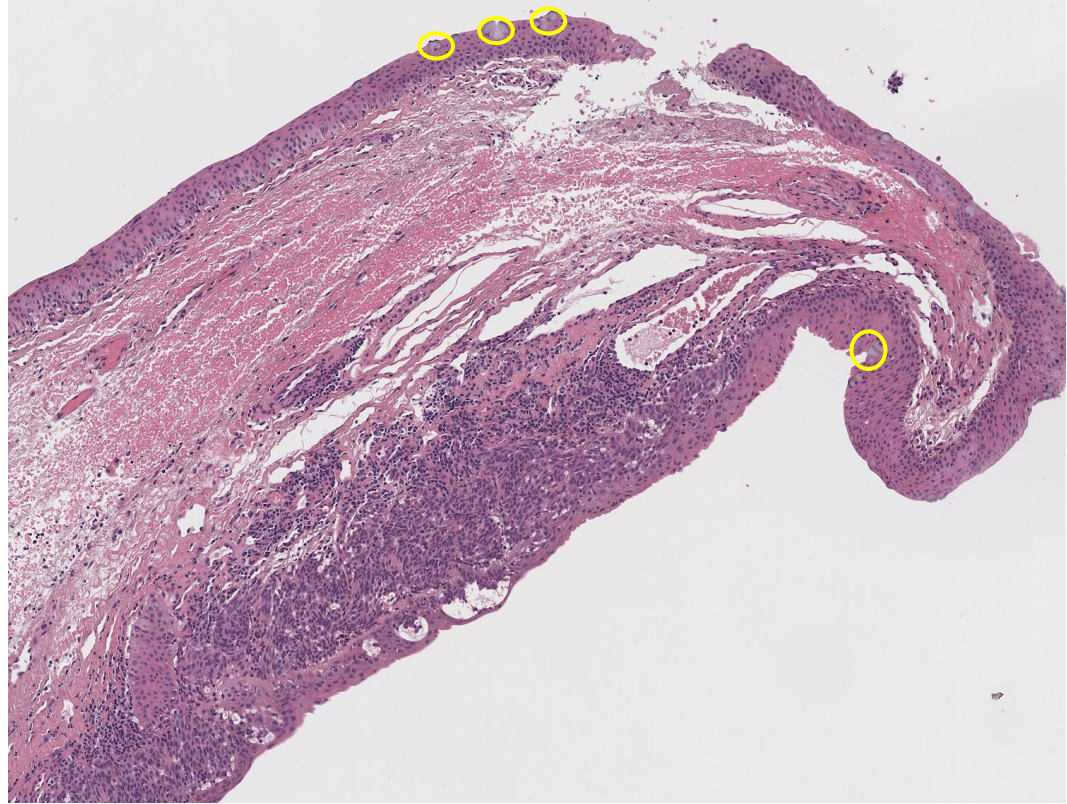
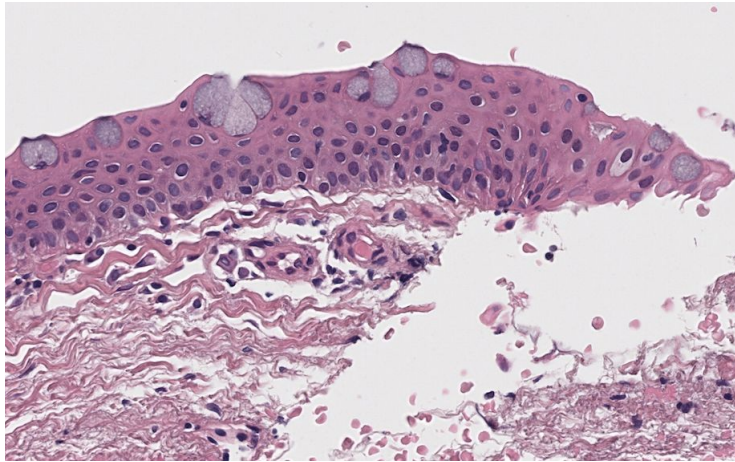
Finding: Some goblet cell bulks clumped next to one another throughout intraepithelial space



Location: Limbus

Case 66 - high-grade
34 year old White male

Finding: Some goblet cell bulks
clumped next to one another
throughout intraepithelial space



2.6 Recommendations

- Consider **stratifying by biological sex**, despite no gender predilection for C-MIL, given the difference in classification accuracy
- Consider **training and testing** both on the **same race** due to 0% correct classification for South Asian and Black individuals
- Even with **low lesion sample sizes** from these regions, it may be worth **stratifying analyses** by region (or otherwise accounting for them) due to differences in cell types and densities
 - a. Higher goblet cell density in plica and caruncle regions than lumbar and bulbar regions, from which the majority of data come from
 - b. Could account for misclassification of benign plica and caruncle slices

2.6 Recommendations Contd.

- Consider a model change to **de-emphasize/ignore goblet-cell-heavy features** (rather than assigning high attention weight to them) as they are largely benign
 - a. High-grade classification accuracy might drop from 100% upon making such a change as some cases may be classified correctly as high-grade for the wrong reasons (e.g. goblet cell bulk presence)
 - i. Inconsistency in goblet cell abundance in benign and high grade slices, reinforcing lack of importance to C-MIL classification

Part 3: Analyst – Visualization

Charlene

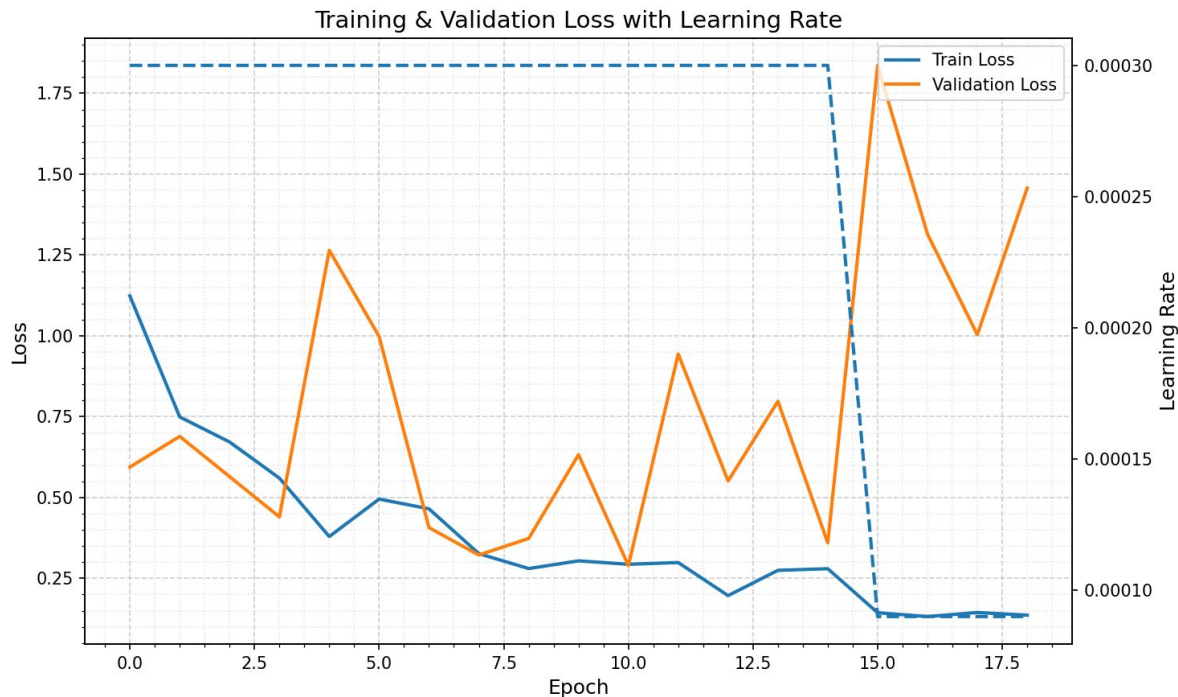
Training and Validation Loss

Training Behavior

- Train loss steadily decreases
- Sharp drop after learning rate decay (~epoch 15)
- Indicates stable optimization

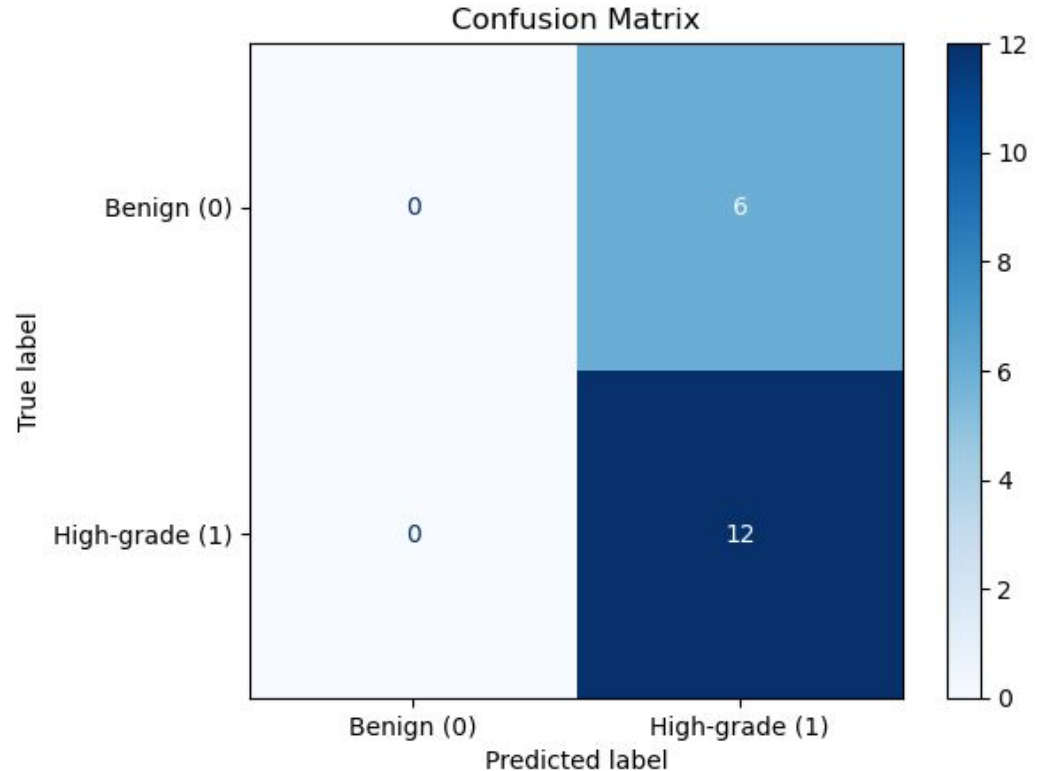
Validation Behavior

- Highly fluctuating validation loss
- No consistent downward trend
- Large spike after learning rate change



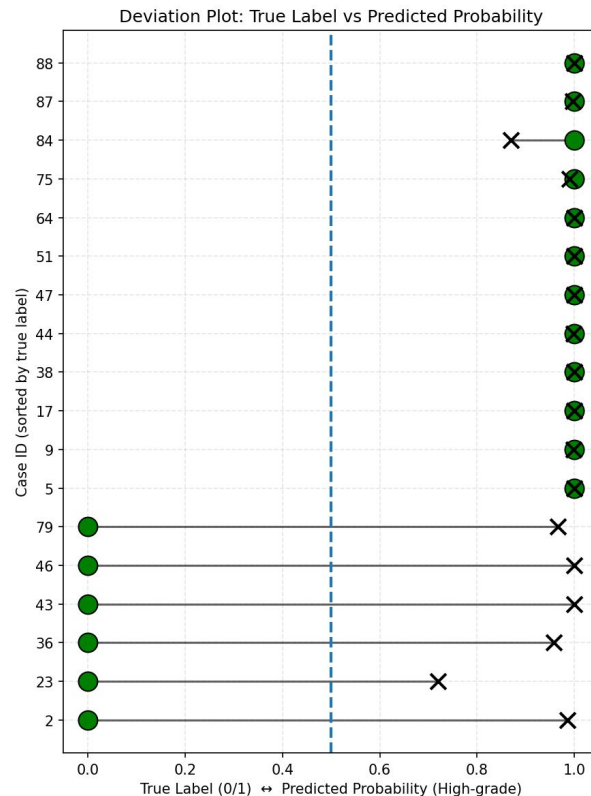
Confusion Matrix

- Accuracy: 66.7%
- Sensitivity (High-grade): 100%
- Specificity (Benign): 0%
- Model predicts all cases as high-grade



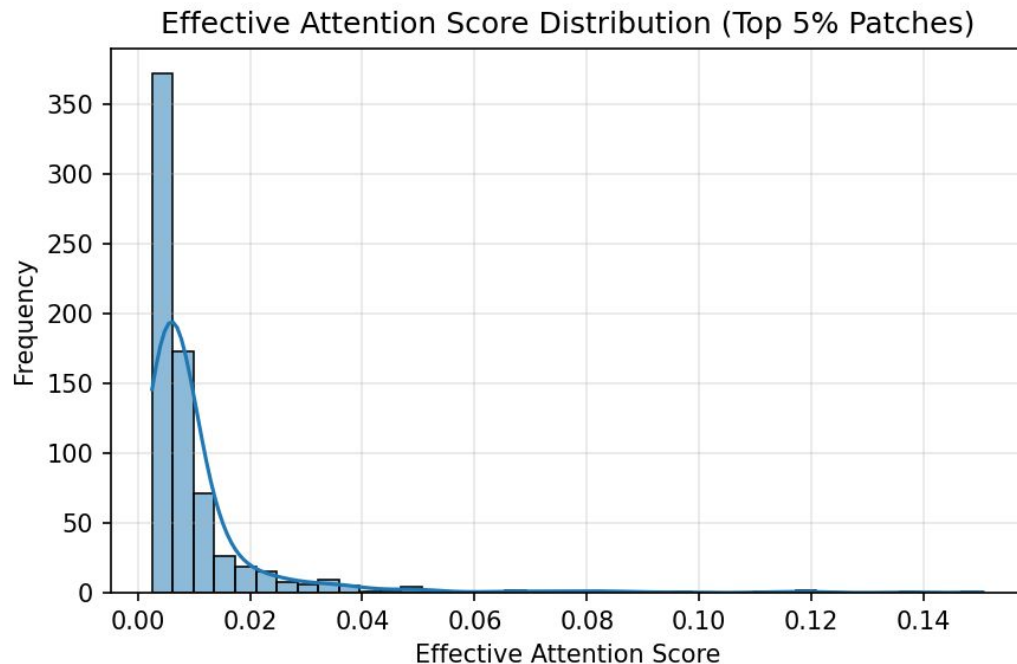
Deviation Plot

- All predictions skewed toward high-grade
- Benign cases misclassified
- High confidence in incorrect predictions
- AUC = 0.736
- Model lacks class separation



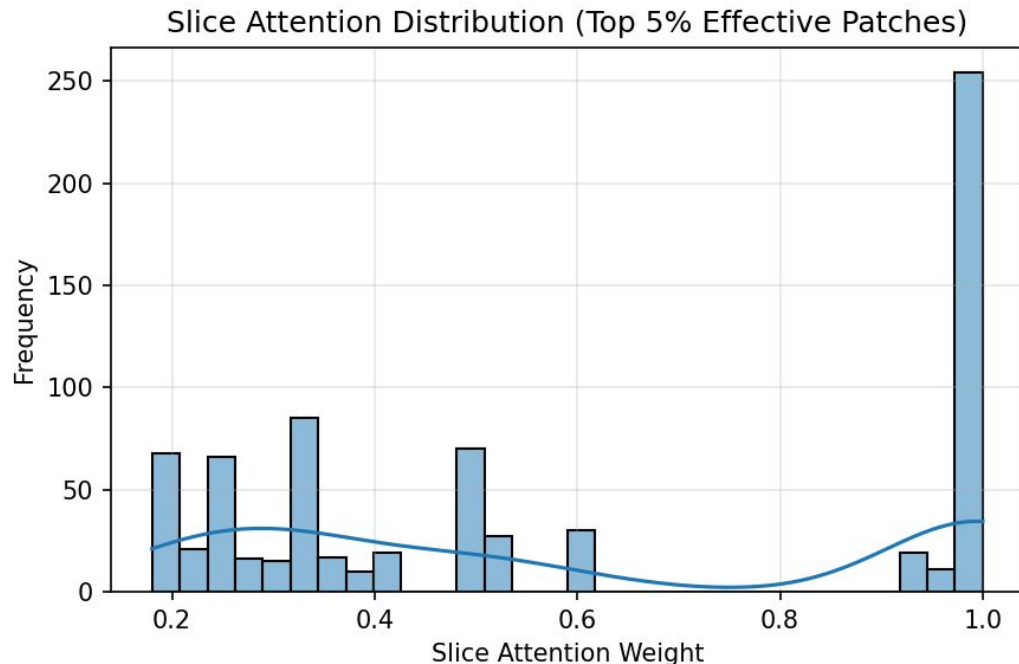
Effective attention score distribution of patches

- Strong right-skewed distribution
- Most scores clustered at low values (~ 0.005 – 0.02)
- Small number of very high-score outliers
- Long tail extending to ~ 0.15

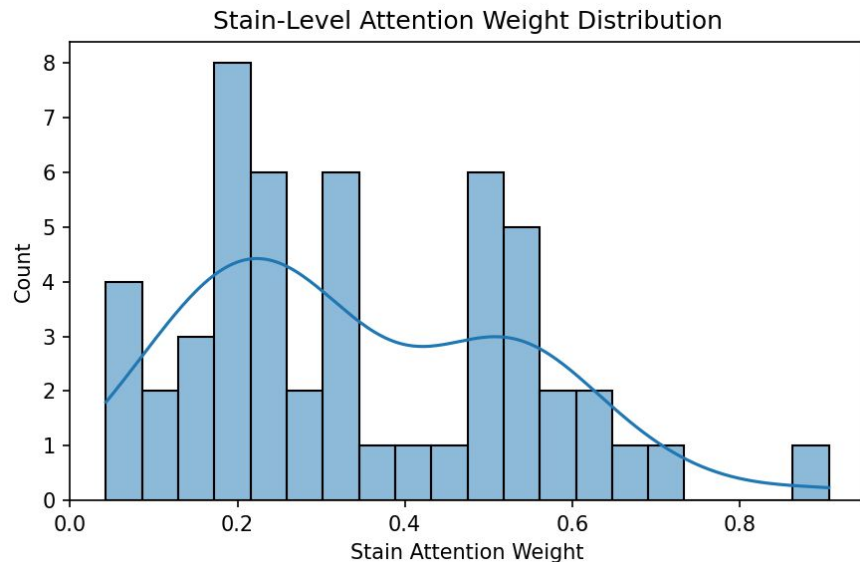


Attention score distribution of slices

- Distribution is multi-modal
- Large concentration near 1.0
- Secondary clusters around 0.2–0.6
- Not smoothly distributed
- Some slices receive near-maximal attention
- Attention across slices is not evenly shared
- Indicates strong slice-level prioritization

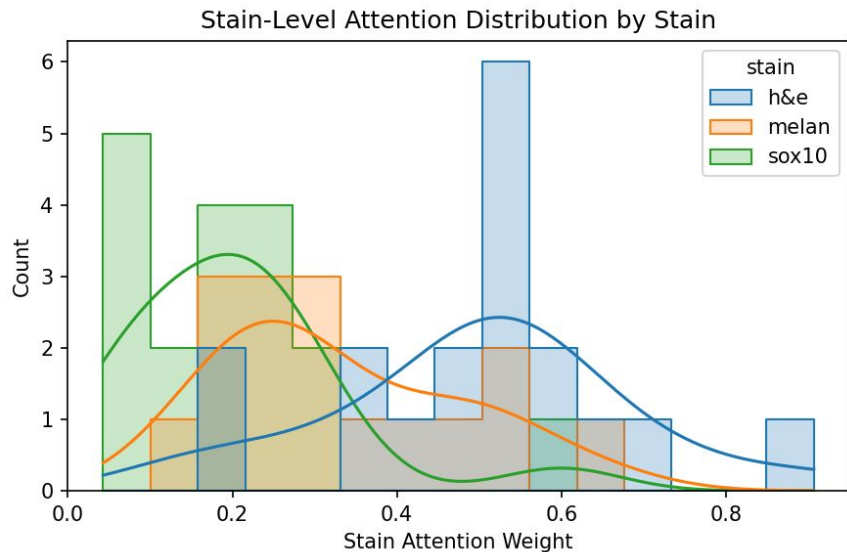


Attention score distribution of stains



- Broad distribution ($\approx 0.05 - 0.9$)
- Majority clustered around $0.15 - 0.4$
- Secondary concentration around $0.45 - 0.6$
- Few high-attention outliers (> 0.7)
- Attention is not uniformly distributed
- Some stains receive substantially higher weight
- Indicates selective stain prioritization

Attention score distribution of stains



H&E

- Wider spread
- Higher maximum values
- Often receives strongest attention

Melan

- Mid-range distribution
- Moderate contribution

SOX10

- Concentrated at lower values
- Less dominant attention

Interpretation

- Clear differentiation in stain importance
- Model learns stain-specific weighting

Top 10 High Attention Patches — Correct High-Grade

case_id	stain	slice_idx	patch_idx	effective_weight
9	melan	0	18	0.136627
51	melan	0	55	0.096442
51	h&e	0	49	0.088196
84	melan	0	160	0.084239
44	melan	1	102	0.079154
44	melan	1	90	0.075936
9	melan	0	5	0.072119
9	melan	0	10	0.060087
17	h&e	1	57	0.052468
17	h&e	2	64	0.051307

Top 10 Low Attention Patches — Correct High-Grade

case_id	stain	slice_idx	patch_idx	effective_weight
64	h&e	1	36	0.002856
64	sox10	1	15	0.002864
64	melan	0	73	0.002866
64	h&e	2	15	0.002872
64	h&e	1	27	0.002874
64	h&e	2	0	0.002876
64	h&e	2	23	0.002882
64	h&e	2	58	0.002886
64	melan	1	12	0.002892
64	h&e	2	24	0.002898

Top 10 High Attention Patches — Incorrect Benign

case_id	stain	slice_idx	patch_idx	effective_weight
46	h&e	0	39	0.150485
46	h&e	0	25	0.120767
46	h&e	0	8	0.117928
46	h&e	0	16	0.112459
46	h&e	0	0	0.081444
46	melan	0	23	0.067403
46	h&e	0	44	0.066922
46	h&e	0	48	0.041740
46	melan	0	32	0.025280
43	h&e	0	42	0.023255

Top 10 Low Attention
Patches — Incorrect
Benign

case_id	stain	slice_idx	patch_idx	effective_weight
79	h&e	2	96	0.002558
79	h&e	1	111	0.002679
79	h&e	2	103	0.002709
79	h&e	0	56	0.002713
79	h&e	1	146	0.002720
79	h&e	0	19	0.002725
79	h&e	1	11	0.002726
79	h&e	0	164	0.002808
79	h&e	0	64	0.002855
79	h&e	2	111	0.002865

Next Steps

Fix Classification Bias

- Apply class-weighted or focal loss
 - Re-evaluate class balance
- Tune decision threshold & calibrate probabilities

Improve Benign Representation

- Add more benign samples or hard negatives
- Review misclassified benign patches

Validate Generalization

- Reassess validation split
- Monitor per-class metrics (recall, specificity)

References

GenAI tools

[Fall 2025 Presentation 4](#)

[Winter 2026 Presentation 6](#)

Butt, K.; Hussain, R.; Coupland, S.E.; Krishna, Y. Conjunctival Melanoma: A Clinical Review and Update. Cancers 2024, 16, 3121. <https://doi.org/10.3390/cancers16183121>

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